GNSS Lab Report - Ri Boss Di Wireless

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ABSTRACT

This report is designed to analyze the reliability and security of the Global Navigation Satellite System (GNSS) on commercial smartphones. Using raw Android measurements, we tested a receiver's behavior in various environments and measured the impact of interference and cyber-spoofing. By analyzing metrics like Carrier-over-Noise density (C/N0) and Horizontal Dilution of Precision (HDOP), this work identified different hardware vulnerabilities and showed the effectiveness of data processing in detecting signal anomalies.

1 INTRODUCTION

Precise positioning is fundamental for everyday location-based services. Although GNSS signals allow for optimized localization, they also expose receivers to attacks such as jamming and spoofing. We analyzed the behavior of signals in multiple scenarios to determine how environmental factors and intentional interference affect the robustness of navigation on consumer devices.

2 METHODOLOGY

Our experimental workflow consisted of field data acquisition and software-based processing of the collected logs.

2.1 Data Acquisition and Scenarios

Raw GNSS data were collected using an Android smartphone, thanks to the *GNSSLogger* analysis app [1]. By activating the "Timed Logging" feature, we ensured exact time observation windows of 300 seconds for each conducted experiment. To evaluate performance limits, measurements were conducted in different environments:

- **Static:** *Open Sky* (baseline)
- **Dynamic:** *Walking* and *Bus* (tracking stability under mobility within an urban canyon).
- **Extreme:** *Aluminium shielding* and *Body interference* (resilience against signal degradation).

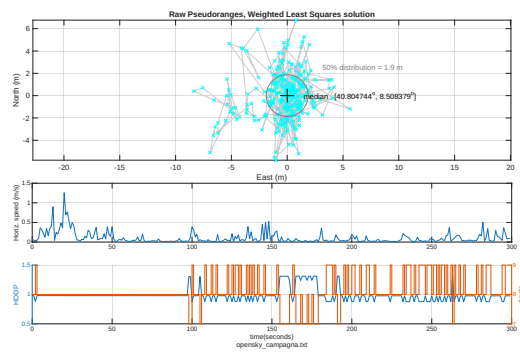
2.2 Analytical Processing and Spoofing

Log files captured with GNSSLogger were processed using the *GNSS Analysis MATLAB Tool* [2] by editing the `ProcessGnssMeasScript` script. This setup allowed for the extraction of positioning and quality metrics, as well as the execution of cyber-spoofing simulations. Specifically, the script emulates a spoofing attack by manipulating the extracted satellite measurements, allowing us to observe how false data shift the calculated coordinates from a set ground-truth.

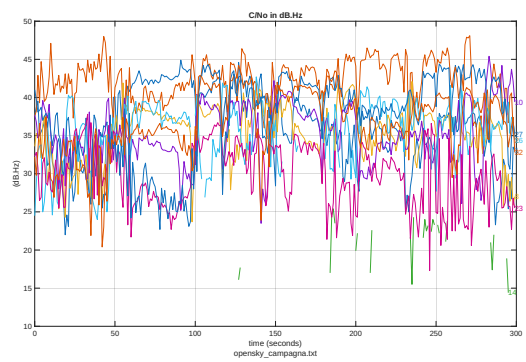
3 MEASUREMENTS

3.1 Open sky

This is the ideal setting, where the receiver is not moving and the satellites are in line-of-sight, ensuring optimal signal reception. The 50% position distribution has a 1.9 m radius (Figure 1a) and the number of satellites remains around 7 to 9 throughout the measurement. The horizontal speed variation is at most around 1.30 m/s (the device was slightly moved during the start of the measurement) and the HDOP, which measures how the satellite geometry affects the accuracy of the horizontal position, remains mostly stable at 1 or lower, spiking only when the satellites drop to 7.



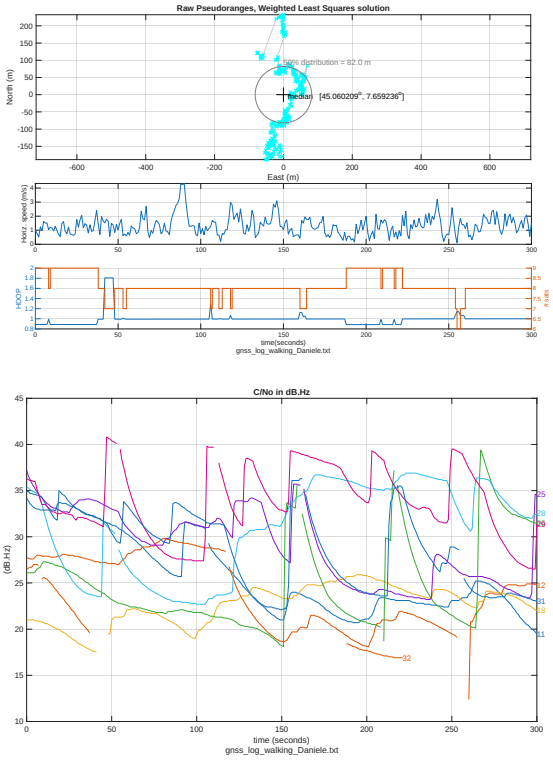
(a) Position estimate, Horizontal speed and HDOP



(b) C/N0

3.2 Outdoor walking

Here, the horizontal speed fluctuates mostly between 0.5 and 2.0 m/s, which is consistent with normal walking. The 4.3 m/s velocity spikes are not possible for a pedestrian, confirming multipath interference at certain intersections and resulting in positioning "jumps". Satellite geometry exhibits noticeable vulnerability: while HDOP is mostly around 0.9-1.0, it sharply peaks at 1.8 when the number of tracked satellites temporarily drops to its minimum of 6 (Figure 2a). The C/N0 plot also confirms the dynamic conditions; while the strongest signals occasionally peak at 40-41 dB-Hz, several satellites experience deep fading, plunging below 15 dB-Hz (e.g., SVID 11 and 32). These frequent signal drops are direct evidence of severe line-of-sight obstructions along the walking path.



(a) Horizontal speed, HDOP and Carrier over Noise

3.3 Outdoor Bus

For this measurement, the receiver is located inside a moving bus, so the line of sight is partially obstructed. Figure 3 shows the receiver's velocity with peaks around 14 m/s (around 50 km/h), which makes sense since the log was registered inside an urban area. The number of visible satellites ranges from 7 to 11 with the corresponding HDOP spikes, probably due to many obstacles around, such as buildings obstructing the line of sight during the trip.

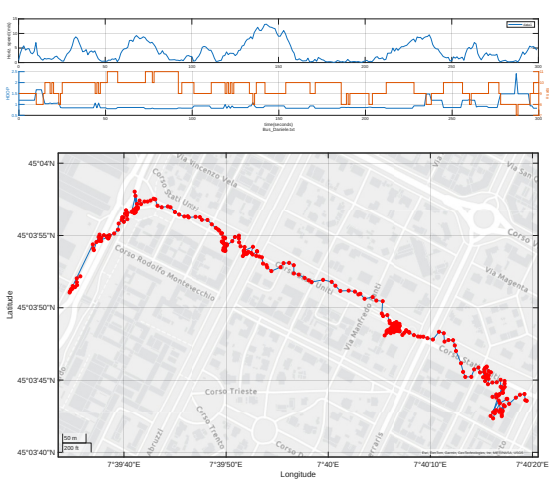
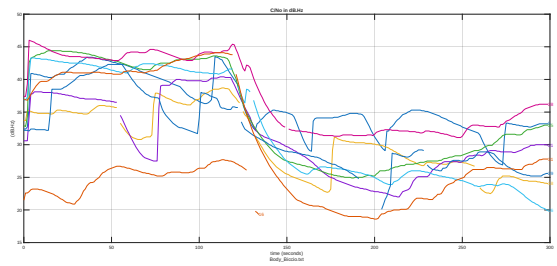


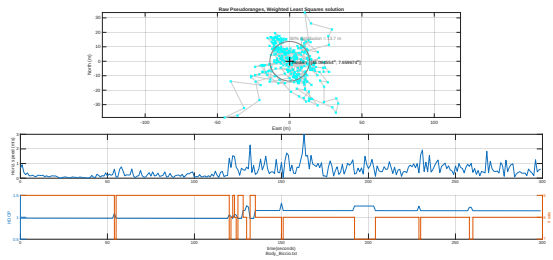
Figure 3: Horizontal speed, HDOP and movement map

3.4 Body interference

The aim of this measurement is to analyze the impact of signal obstruction caused by the human body. This happens because our body absorbs radio waves in the GNSS L-bands, acting as an attenuating shield that obstructs signal reception. The 5-minute recording is divided into two parts: in the first 2 minutes the receiver is in normal conditions, and in the next 3 minutes the device is on the chest of the operator, covered by his hands. The first indicator of the scenario change is the C/N0 graph (Figure 4a) where levels during the initial 120 seconds are steady and comparatively high (35/45 dB-Hz), indicative of clean reception. The strength of the signals drops to about 10-15 dB-Hz when the phone is placed near the chest: this interference is clearly visible in the three panels of Figure 4b, where the number of satellites (orange line) after 120 seconds becomes unstable along with tracking losses. As a result, the HDOP (blue line) increases and reaches a peak. When some satellites are lost, the geometry of the visible constellation is compromised, resulting in deterioration of accurate positioning. Although the receiver remained stationary throughout the test, the velocity plot shows significant noise after 120 seconds, almost certainly due to low C/N0 and multipath induced by body proximity. In the scatter plot, the dispersion of the points increases consistently in the second phase: while the first points are concentrated near the origin, the next ones suggest systematic errors resulting from interference.



(a) C/N0 in dB.Hz

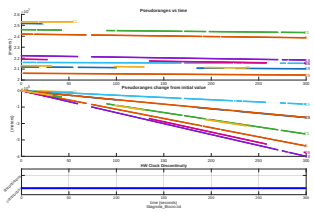


(b) Position estimate, Horizontal speed and HDOP

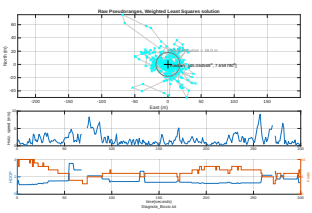
3.5 Aluminum Interference

In this experiment, we simulated a Faraday cage using a sheet of aluminum foil, leaving a small opening to allow a portion of the satellite signals to pass through. The C/N_0 analysis in Figure 5c confirms that placing our smartphone inside an aluminum foil caused an immediate and severe attenuation of the signals. Satellites 4 and 11 disappeared, and other signals' power decreased significantly with levels between 20 and 35 dB-Hz. On the other hand, signals from satellites 5 to 29 showed relative stability or intermittent peaks,

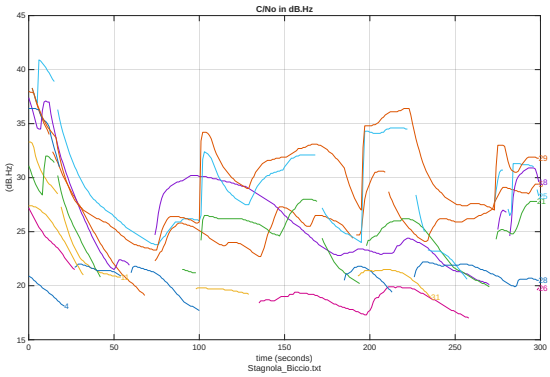
proving that these satellites were periodically aligned with the uncovered portion of the foil. The signal instability is clearly visible in Figure 5a but the clock remained stable. Figure 5b reveals a significant spread of data points, characterized by a 50% distribution error of 18.0 meters from the median position. Even though the smartphone was fixed in place, the horizontal speed graph displays anomalous velocity peaks, reaching up to 10 m/s due to phase noise and multipath reflections caused by the metal foil. This instability is also reflected in the satellite geometry, as the HDOP value lies between 1.5 and 2.5 due to satellites frequently being lost and reacquired through the hole. The experiment demonstrates that even partial shielding is enough to compromise positioning accuracy, even though hardware clock continuity was maintained throughout the test.



(a) Pseudorange & HW Clock



(b) Position Error, HDOP & N. Sat



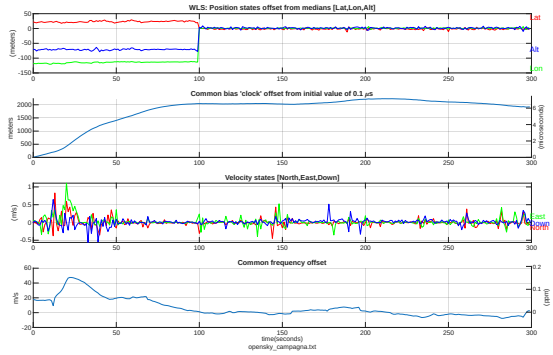
(c) C/N0

4 SPOOFING

This section will explore the impact of simulated cyber-spoofing attacks, which were performed by manipulating the previously captured open sky measurement. We triggered different attack conditions starting at $t = 100$ s to observe how the receiver handles spatial (and also temporal) anomalies.

4.1 Position Spoofing near receiver (without delay)

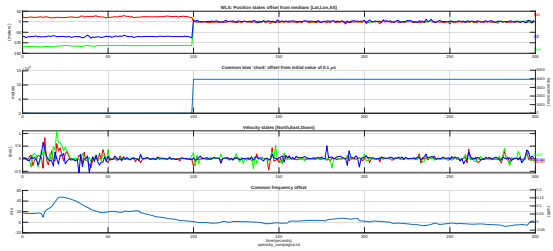
In this scenario, we simulated a spoofing attack by altering the `spoof.position` coordinates without injecting any delay. When observing the pseudorange variation over time, the attack is virtually imperceptible: because the geometric shift induced by the spoofing is relatively small (around 100-150 meters) no obvious step is immediately visible. However, the attack is easily exposed by analyzing the Weighted Least Squares (WLS) solver, which takes these manipulated measurements and causes an instantaneous change in latitude, longitude, and altitude values. Given that no artificial delay was introduced, the receiver's clock bias remains perfectly continuous and unaffected by the spoofing injection: the algorithm interprets the pseudorange jumps simply as an instantaneous physical translation of the receiver in space, validating the "stealthiness" of a purely geometric spoofing attack.



(a) Position states and Clock bias

4.2 Position Spoofing near receiver (with delay)

In this second scenario, we injected both a false spatial position and a specific `spoof.delay` of 4 ms: this new addition creates a massive vertical step across all pseudoranges simultaneously. Because satellite signals travel at the speed of light, even a fractional time delay introduced translates into hundreds of kilometers of artificial distance added to every measurement. The WLS solver processes this new measurement and interprets it as a massive receiver clock error. As shown in Figure 7a, while the spatial coordinates (latitude, longitude, altitude) still snap to the fake location, the clock bias experiences an instant and drastic jump exactly at 100 seconds. This behavior highlights a critical trade-off for an attacker: while introducing a delay might be physically necessary to "capture" the target receiver in a real-world over-the-air attack, it leaves an obvious anomaly in the clock bias domain.

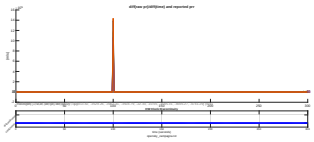


(a) Position states and Clock bias jump

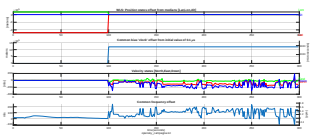
4.3 Position Spoofing far from receiver (with delay)

Here we executed a harsh spoofing experiment, shifting the target coordinates by approximately 478 km while simultaneously injecting a time delay. This attack represents a worst-case scenario where both the spatial and temporal domains are severely compromised. The "aggressive" spatial translation is visible in the pseudorange rates (Figure 8a): in the WLS solution plot the offsets for latitude,

longitude and altitude immediately jump on the order of 10^5 meters. Consequently, the WLS solver is forced to compensate for this artificial propagation delay by forcefully adjusting the receiver's clock. As observed in the previous delayed scenario, the common bias clock violently spikes as shown in Figure 8b.



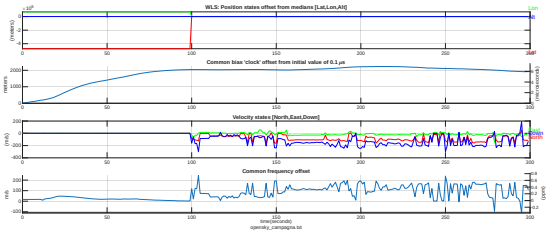
(a) Pseudorange rates anomaly



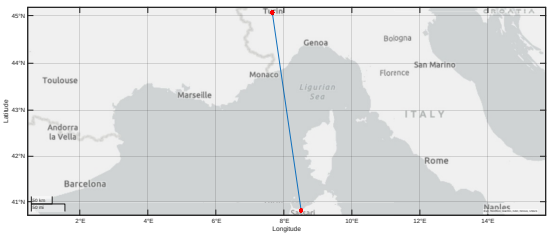
(b) Position and Clock bias jump

4.4 Position Spoofing far from receiver (without delay)

Lastly, we simulated another attack by forcing the coordinates similarly to the previous experiment but with the `spoof.delay` parameter disabled. The magnitude of the spatial jump forces the WLS solver to register again a violent discontinuity in the position states (Figure 9a), with offsets instantly surging by an order of 10^5 meters and generating a massive change in the pseudorange rates. But the most revealing metric here is the receiver's internal clock: since no artificial time delay was introduced, the common bias clock smoothly follows its natural continuous drift. This physical behavior confirms that the WLS solver interpreted the manipulated measurements strictly as an instantaneous spatial teleportation, compromising the position domain while leaving the temporal synchronization intact.



(a) Position jump of $\times 10^5$ m with continuous clock



(b) Actual location vs spoofed location

4.5 Considerations about Spoofing

The simulation results fully matched our expectations: the pure spatial manipulation successfully altered the physical coordinates in the WLS solver, while the addition of a temporal delay predictably manifested an extreme spike in the hardware clock bias. A reliable spoofing detection strategy could easily track these metrics, flagging physically impossible discontinuities in the pseudorange rates or in the clock bias. From an attacker's perspective, to make the software attack even stealthier, the code could be modified to inject the `spoof.position` and `spoof.delay` progressively over time, rather than an instantaneous step: this would keep the horizontal velocity and clock drift within physically plausible limits, with more chances of bypassing basic anomaly detection algorithms. Furthermore, it is important to note a key characteristic of this purely software-based attack: because the script manipulates the data and does not synthesize a raw signal, the C/N_0 plots show absolutely no difference during the attack. In a "real-world" spoofing event, the attacker must also transmit fake signals at a higher power to capture the receiver's tracking loops, generating abrupt variations in the C/N_0 levels. Although our data-level injection bypasses these physical-layer checks, monitoring signal strength anomalies remains a primary real-world defense mechanism.

5 RESULTS AND CONCLUSION

This study highlighted several vulnerabilities of consumer-grade GNSS receivers. Table 1 summarizes the overall performance metrics, confirming the Open Sky scenario as the ideal one for optimal reception. In contrast, introducing electromagnetic shielding and human proximity (during the Aluminum and Body trials) induced drastic attenuation and compromised tracking. The "dynamic" tests also mirrored these challenges: while the moving Bus retained an adequate constellation layout despite reduced power, the Walking path suffered from an "urban canyon" effect, leading to heavy fluctuations and sharp HDOP peaks. We also turned on battery-saving mode multiple times during these experiments, but it showed no predictable (or even significant) impact. Furthermore, our cyber-spoofing simulations proved the ease of manipulating spatial coordinates; although limited by the absence of raw RF transmission, these attacks still left detectable traces. Finally, the collected data confirm that commercial navigation remains highly sensitive to a combination of environmental blockages, multipath interference and software-level attacks.

Table 1: Signal Quality and Satellite Geometry

Scenario	C/N0 (dB-Hz)		HDOP	
	Mean	Std. Dev.	Mean	Std. Dev.
Open Sky	35.88	5.44	1.00	0.10
Outdoor Walking	27.91	5.18	1.09	0.23
Outdoor Bus	27.08	3.58	1.05	0.25
Body Interference	32.96	6.94	1.10	0.12
Aluminum Shielding	26.67	4.37	2.08	0.65

REFERENCES

[1] Google Inc. 2016. GPS logger. (2016). github.com/google/gps-measurement-tools/releases/tag/2.0.0.1
[2] Google Inc. 2016. GPS Measurement Tools. (2016). github.com/google/gps-measurement-tools